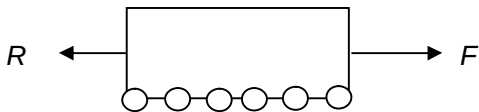


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$$P = Fv$$

$$F = \frac{2 \times 10^6}{40} = 5 \times 10^4 \text{ N. Hence } R = 5 \times 10^4 \text{ N}$$

At a lower speed of 10 m s^{-1} , the force $F' = \frac{2 \times 10^6}{10} = 20 \times 10^4 \text{ N}$.

$$\text{Resultant force} = (20 - 5) \times 10^4 = 1.5 \times 10^5 \text{ N}$$

Answer: B

$$10 \text{ Loss in gravitational PE} = mgh = 4.2 \times 9.81 \times 2.9 = 12 \text{ J.}$$